

ANALYSIS REPORT

January 18th 2024

Your reference: [REDACTED] | Basalt Eifelgold <2mm 2022 | [REDACTED]

Our reference: 2302AZ_EN_v3

Table 1: Sample list with requested analyses.

Our reference	Your reference	XRF	XRD	PSD	BET
[REDACTED]	[REDACTED]				
2302AZ02	Basalt Eifelgold <2mm 2022	X	X	X	X
[REDACTED]	[REDACTED]				

Table 2: Measurement characteristics of the bulk XRD analysis.

Parameter	Value
Analysis	Bulk mineralogical analysis by X-ray diffraction
General sample handling	Drying at 40°C, homogenisation by mortar & pestle
Sample preparation	Wet milling and drying
Diffractometer	Bruker D8 Advance, XE-T detector, Cu-K α radiation
Data treatment methodology	In-house
Interpretation by	[REDACTED]
Date of measurement	06-07/03/2023
Date of data treatment	11/04/2023
Results	Table 6 and Figures 1-4

Your reference:

Our reference:

Table 3: Measurement characteristics of the XRF analysis.

Parameter	Value
Analysis	X-ray Fluorescence analysis
Sample preparation	Prior to fusion, the loss on ignition (LOI), which includes H ₂ O, CO ₂ , S and other volatiles, is determined from the weight loss after roasting the sample at 1000°C for 2 hours. The fusion disk is made by mixing a 0.75 g equivalent of the roasted sample with 9.75 g of a combination of lithium metaborate and lithium tetraborate with lithium bromide as a releasing agent. The samples were fused in a Pt crucible using an automated crucible fluxer and automatically poured into a Pt mold for casting.
Apparatus	Panalytical Axios Advanced wavelength dispersive XRF
Date of measurement	21/03/2023
Results	Table 7

Table 4: Characteristics of the grain-size analysis.

Parameter	Value
Analysis	Laser diffraction
Sample preparation	Dispersion in water and splitting the sample – individual splits are added to the instrument until a proper obscuration is obtained
Apparatus	Fritsch Analysette A-22 NeXt
Use of ultrasonic dispersion	Ja
Model	Mie theory
Measured by	A.A.
Date of analysis	20/03/2023
Results	Table 8-9 and Figure 5

Table 5: Measurement characteristics of the BET analysis.

Parameter	Value
Instrument	Quantachrome Autosorb
Date of analysis	04/06/2021 – 08/06/2021
Date of data treatment	08/06/2021
Operator	A.A.
Sample preparation	Outgassing 2h at 200°C under high-vacuum
Adsorptive-gas	N ₂
Temperature during analysis	77.35K (liquid nitrogen)
Results	Table 10 and Figure 6

Your reference: XXXXXXXXXX Basalt Eifelgold <2mm 2022 | XXXXXXXXXX
 Our reference: 2302AZ_EN_v3

Table 6: Quantitative bulk mineralogical compositions of the samples (in weight percentages of the identified crystalline components).

Mineral	Mineral Formula ¹	Basalt Eifelgold <2mm 2022
SILICATES		
Quartz	SiO ₂	
K-feldspar	KAlSi ₃ O ₈	
Plagioclase	(Na,Ca)(Si,Al) ₄ O ₈	7,7
Clinopyroxene	(Ca,Na)(Mg,Fe,Al,Ti)(Si,Al) ₂ O ₆	49,4
Orthopyroxene	(Ca,Na)(Mg,Fe,Al,Ti)(Si,Al) ₂ O ₆	
Amphibole	Ca ₂ (Mg,Fe) ₅ Si ₈ O ₂₂ (OH) ₂	
Olivine	(Mg,Fe) ₂ SiO ₄	11,2
Nepheline	(Na,K)AlSiO ₄	9,3
Sodalite	Na ₈ Al ₆ Si ₆ O ₂₄ Cl ₂	0,5
Leucite	KAlSi ₂ O ₆	12,6
Analcime	NaAlSi ₂ O ₆ •(H ₂ O)	1,3
Alpha-Dicalcium Silicate hydrate	α-Ca ₂ (SiO ₄)(H ₂ O)	
Beta C2S (Larnite)	Ca ₂ SiO ₄	
Alpha' C2S	Ca ₂ SiO ₄	
2:1 layer silicates	(K,H ₃ O)(Al,Mg,Fe) ₂ (Si,Al) ₄ O ₁₀ [(OH) ₂ , (H ₂ O)]	1,8
Chlorite	(Mg,Fe) ₅ Al(Si ₃ Al)O ₁₀ (OH) ₈	
Garnet	X ₃ Z ₂ (SiO ₄) ₃	
Serpentine	(Mg,Fe,Ni,Mn,Al,Zn) ₃ [Si ₂ O ₅](OH) ₄	
Titanite	CaTiSiO ₅	
CARBONATES		
Calcite	CaCO ₃	
PHOSPHATES		
Apatite	Ca ₅ (PO ₄) ₃ (OH,F,Cl)	1,2
OXIDES/HYDROXIDES		
Anatase	TiO ₂	
Hematite	Fe ₂ O ₃	
Spinel-type minerals	Fe ²⁺ Fe ³⁺ ₂ O ₄	4,8
Wuestite	FeO	
Lime	CaO	
Portlandite	Ca(OH) ₂	
Periclase	MgO	0,7

¹ These formulae are general formulae and do not necessarily correspond to the composition of the minerals in these specific samples.

Your reference: [REDACTED] Basalt Eifelgold <2mm 2022 | [REDACTED]
 Our reference: Z302AZ_EN_v3

Srebrodolskite	Ca₂Fe³⁺₂O₅	17,8
ELEMENTS		
Iron	Fe	0,7

Table 7: Results of the XRF analyses. The Loss on Ignition (LOI) was determined on the fused samples.

Element	Unit	Detection limit	Basalt Eifelgold <2mm 2022
SiO ₂	%	0.01	43,3
Al ₂ O ₃	%	0.01	13,4
Fe ₂ O ₃	%	0.01	10,95
MnO	%	0.01	0,185
MgO	%	0.01	9,8
CaO	%	0.01	12,7
Na ₂ O	%	0.01	2,63
K ₂ O	%	0.01	3,08
TiO ₂	%	0.01	2,73
P ₂ O ₅	%	0.01	0,55
LOI	%		0,6
TOTAL	%		100
Cr	ppm	68	205
V	ppm	17	286
Co	ppm	37	51
Cu	ppm	40	80
Ni	ppm	24	118

Table 8: Laser diffraction: raw data (cumulative frequency distributions, in vol%).

Size (µm)	Basalt Eifelgold <2mm 2022	Size (µm)	Basalt Eifelgold <2mm 2022	Size (µm)	Basalt Eifelgold <2mm 2022
0,011	0,0	1,38	0,6	170.35	65,1
0,012	0,0	1,53	1,0	189.60	65,4
0,014	0,0	1,71	1,5	211.02	66,2
0,015	0,0	1,90	2,1	234.87	67,5
0,017	0,0	2,11	2,9	261.41	69,6
0,019	0,0	2,35	3,8	290.96	72,6
0,021	0,0	2,62	4,9	323.84	76,4
0,024	0,0	2,91	6,1	360.43	80,8
0,026	0,0	3,24	7,4	401.16	85,5
0,029	0,0	3,61	8,8	446.50	89,9
0,032	0,0	4,02	10,3	496.96	93,8
0,036	0,0	4,47	11,8	553.12	96,7
0,040	0,0	4,98	13,4	615.62	98,5
0,045	0,0	5,54	15,0	685.19	99,5
0,050	0,0	6,16	16,6	762.62	99,9

0,055		0,0	6,86		18,2	848.81		00,0
0,062		0,0	7,64		19,8	944.73		00,0
0,069		0,0	8,50		21,5	1051.49		00,0
0,076		0,0	9,46		23,1	1170.32		00,0
0,085		0,0	10,53		24,8	1302.57		00,0
0,095		0,0	11,72		26,6	1449.77		00,0
0,105		0,0	13,04		28,4	1613.61		00,0
0,117		0,0	14,52		30,4	1795.96		00,0
0,131		0,0	16,16		32,4	1998.92		00,0
0,145		0,0	17,98		34,5			
0,162		0,0	20,02		36,7			
0,180		0,0	22,28		38,9			
0,200		0,0	24,80		41,1			
0,223		0,0	27,60		43,3			
0,248		0,0	30,72		45,5			
0,276		0,0	34,19		47,7			
0,308		0,0	38,05		49,8			
0,342		0,0	42,35		51,8			
0,381		0,0	47,14		53,8			
0,424		0,0	52,46		55,8			
0,472		0,0	58,39		57,6			
0,525		0,0	64,99		59,4			
0,585		0,0	72,34		60,9			
0,651		0,0	80,51		62,2			
0,724		0,0	89,61		63,3			
0,806		0,0	99,73		64,0			
0,897		0,0	111,01		64,5			
1,00		0,1	123,55		64,8			
1,11		0,2	137,51		64,9			
1,24		0,4	153,05		65,0			

Table 8: Continued.

Case	Case	Case	Case
1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16
17	18	19	20
21	22	23	24
25	26	27	28
29	30	31	32
33	34	35	36
37	38	39	40
41	42	43	44
45	46	47	48
49	50	51	52
53	54	55	56
57	58	59	60
61	62	63	64
65	66	67	68
69	70	71	72
73	74	75	76
77	78	79	80
81	82	83	84
85	86	87	88
89	90	91	92
93	94	95	96
97	98	99	100

Your reference: Steelslag | Basalt Eifelgold <2mm 2022 | "Olivine" Sibelco (NO) | Diabase Schicker
 Our reference: 2302AZ_EN_v3



Table 9: Statistical parameters of the grain-size distribution of the sample (in μm).

Parameter	Basalt Eifelgold <2mm 2022		
Mean	152,7		
Standard Deviation	188,5		
Mode	401,2		
d10 (μm)	3,9		
d50 (μm)	38,5		
d60 (μm)	68,1		
d90 (μm)	447,5		

Table 10: BET specific surface area: results.

Sample	BET area (m^2/g)	Correlation coefficient BET plot	Sample mass (g)
Basalt Eifelgold <2mm 2022	2.3	0.999973	1.6070

Your reference
Our reference

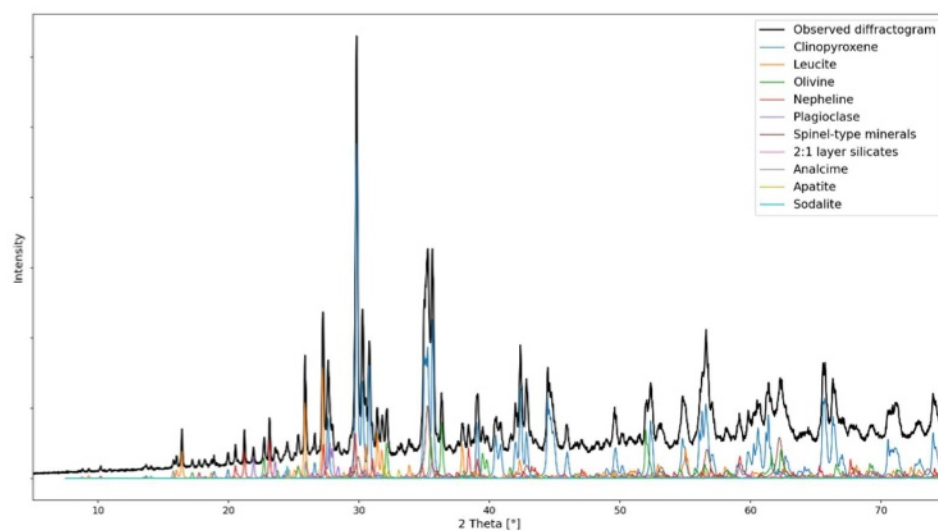
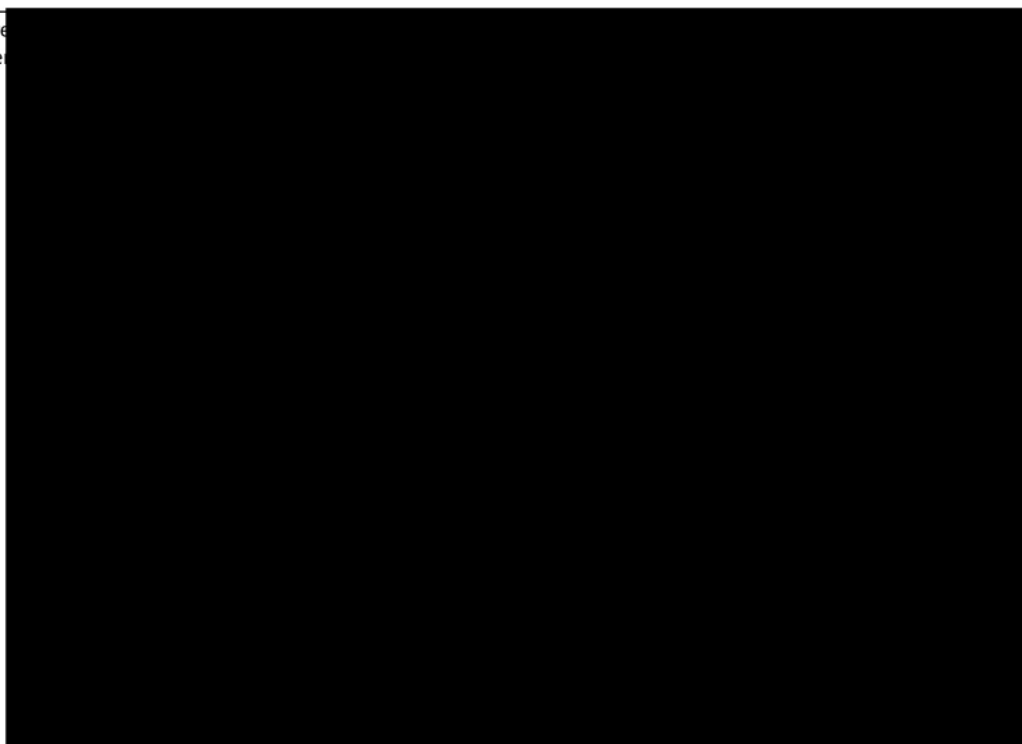


Figure 2: Diffraction pattern of the sample “Basalt Eifelgold <2mm 2022”. The minerals that contribute to the main reflections are indicated.



